

■ 1.7.2 Making Tables of Values

You can use lists as tables of values. You can generate the tables, for example, by evaluating an expression for a sequence of different parameter values.

This gives a table of the values of i^2 , with i running from 1 to 6.

```
In[1]:= Table[i^2, {i, 6}]
Out[1]= {1, 4, 9, 16, 25, 36}
```

Here is a table of $\sin(n\pi/5)$ for n from 0 to 4.

```
In[2]:= Table[Sin[n Pi/5], {n, 0, 4}]
Out[2]= {0, Sin[ $\frac{\text{Pi}}{5}$ ], Sin[ $\frac{2 \text{ Pi}}{5}$ ], Sin[ $\frac{3 \text{ Pi}}{5}$ ], Sin[ $\frac{4 \text{ Pi}}{5}$ ]}
```

This gives the numerical values.

```
In[3]:= N[%]
Out[3]= {0., 0.587785, 0.951057, 0.951057, 0.587785}
```

You can also make tables of formulae.

```
In[4]:= Table[x^i + 2i, {i, 5}]
Out[4]= {2 + x, 4 + x^2, 6 + x^3, 8 + x^4, 10 + x^5}
```

`Table` uses exactly the same “iterator” notation as the functions `Sum` and `Product`, which were discussed in Section 1.5.4.

```
In[5]:= Product[x^i + 2i, {i, 5}]
Out[5]= (2 + x) (4 + x^2) (6 + x^3) (8 + x^4) (10 + x^5)
```

This makes a table with values of x running from 0 to 1 in steps of 0.25.

```
In[6]:= Table[Sqrt[x], {x, 0, 1, 0.25}]
Out[6]= {0, 0.5, 0.707107, 0.866025, 1.}
```

You can perform other operations on the lists you get from `Table`.

```
In[7]:= %^2 + 3
Out[7]= {3, 3.25, 3.5, 3.75, 4.}
```

`TableForm` displays lists in a “tabular” format. Notice that both the words in the name `TableForm` begin with capital letters.

```
In[8]:= % // TableForm
          3
          3.25
          3.5
          3.75
Out[8]//TableForm= 4.
```

All the examples so far have been of tables obtained by varying a single parameter. You can also make tables that involve several parameters. These “multidimensional” tables are specified using the standard *Mathematica* iterator notation, discussed in Section 1.5.4.

This makes a table of $x^i + y^j$ with i running from 1 to 3 and j running from 1 to 2.

```
In[9]:= Table[x^i + y^j, {i, 3}, {j, 2}]
Out[9]= {{x + y, x + y^2}, {x^2 + y, x^2 + y^2},
          {x^3 + y, x^3 + y^2}}
```

The table in this example is a *list of lists*. The elements of the outer list correspond to successive values of i . The elements of each inner list correspond to successive values of j , with i fixed.

Sometimes you may want to generate a table by evaluating a particular expression many times, without incrementing any variables.

This creates a list containing four copies of the symbol x .

```
In[10]:= Table[x, {4}]
Out[10]= {x, x, x, x}
```

This gives a list of four pseudorandom numbers. Table re-evaluates Random[] for each element in the list, so that you get a different pseudorandom number.

```
In[11]:= Table[Random[], {4}]
Out[11]= {0.353297, 0.0205034, 0.117374, 0.546051}
```

Table[f, {imax}]	give a list of $imax$ values of f
Table[f, {i, imax}]	give a list of the values of f as i runs from 1 to $imax$
Table[f, {i, imin, imax}]	give a list of values with i running from $imin$ to $imax$
Table[f, {i, imin, imax, istep}]	use steps of $istep$
Table[f, {i, imin, imax}, {j, jmin, jmax}, ...]	generate a multi-dimensional table
TableForm[list]	display a list in tabular form

Functions for generating tables.

You can use the operations discussed in Section 1.2.3 to extract elements of the table.

This creates a 2×2 table, and gives it the name m .

```
In[12]:= m = Table[i - j, {i, 2}, {j, 2}]
Out[12]= {{0, -1}, {1, 0}}
```

This extracts the first sublist from the list of lists that makes up the table.

```
In[13]:= m[[1]]
Out[13]= {0, -1}
```

This extracts the second element of that sublist.

```
In[14]:= %[[2]]
Out[14]= -1
```

This does the two operations together.

```
In[15]:= m[[1,2]]
Out[15]= -1
```

This displays `m` in a “tabular” form.

```
In[16]:= TableForm[m]
          0      -1
Out[16]//TableForm= 1      0
```

$t[[i]]$	give the i^{th} sublist in t
$t[[\{i_1, i_2, \dots\}]]$	give a list of the $i_1^{\text{th}}, i_2^{\text{th}}, \dots$ parts of t
$t[[i, j, \dots]]$	give the part of t corresponding to $t[[i]][[j]]\dots$

Ways to extract parts of tables.

As we mentioned in Section 1.2.3, you can think of lists in *Mathematica* as being analogous to “arrays”. Lists of lists are then like two-dimensional arrays. When you lay them out in a tabular form, the two indices of each element are like its x and y coordinates.

You can use `Table` to generate arrays with any number of dimensions.

This generates a three-dimensional $2 \times 2 \times 2$ array. It is a list of lists of lists.

```
In[17]:= Table[i j^2 k^3, {i, 2}, {j, 2}, {k, 2}]
Out[17]= {{1, 8}, {4, 32}}, {{2, 16}, {8, 64}}}
```